Penn Engineering
University of Pennsylvania

FA26

Special Seminar on
Multi-Modal AI

Jiayuan Mao

Department of CIS, GRASP

https://cis7000-fall26.thelabone.org/

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Lecture 16: Scaling and Multimodal Data

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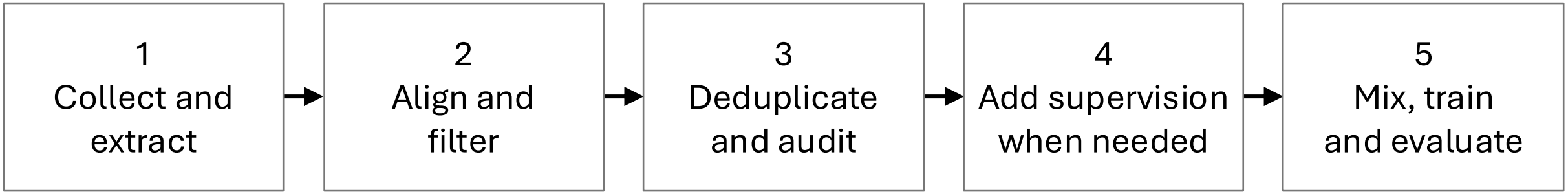


Overview: A day as a “frontier-lab” researcher

Part	Main question
1 Data pipeline	How do raw sources become useful training examples?
2 Scaling laws	How should we use a fixed compute budget?

The multimodal data pipeline

Goal: produce training examples that support a target task



Example



A: caliper brake



B: cantilever brake

Stage	What changes in this example
1 Extract	Keep the photos, figure captions and page URL
2 Align + filter	6 candidates become 4 usable images Check caption assignments against the page
3 Deduplicate	4 images become 2 distinct photos Remove the repeated download and thumbnail
4 Add supervision	Optional next step: write and verify repair Q&A
5 Train + evaluate	Choose the sampling share, then test brake recognition

Stages 1–3: worked example Stages 4–5: design choices for the assistant

https://en.wikipedia.org/wiki/Bicycle_brake

Sources of multimodal training data



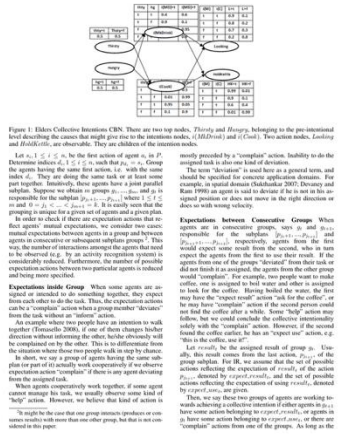
Web image + alt text

Heavenly Tea Leaves



Webpage with image + nearby text

Tales of the Travelbug



PDF + figure

AAAI paper, page 4



Image + instruction

COCO image

LAION: an image and its actual alt text

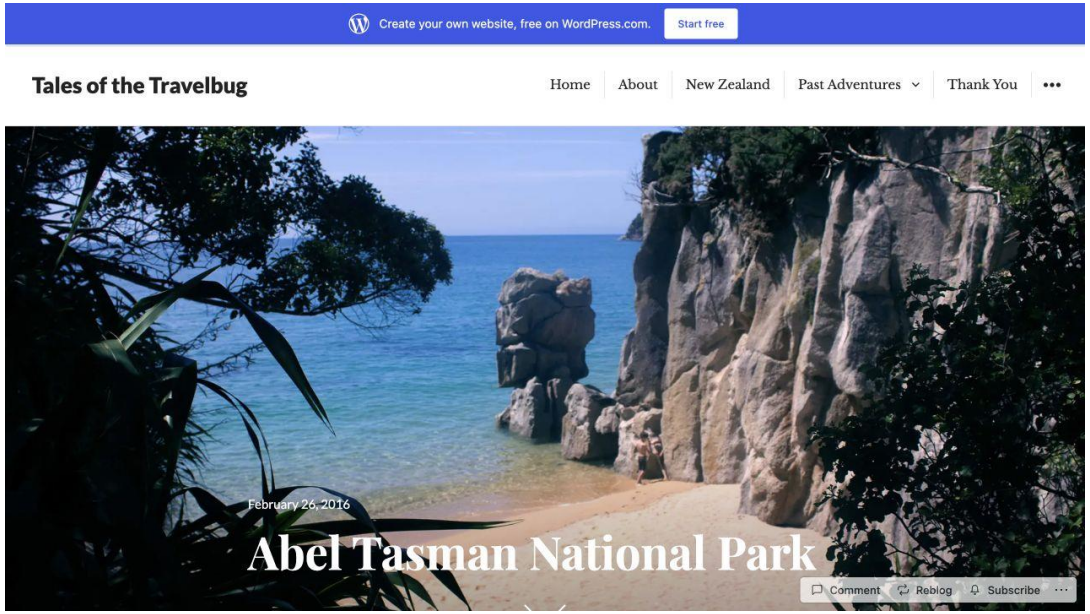


Fields in the released web metadata

Field	Actual value
alt_text	Exploring the Green Teas of Japan
filename	Japanese_Tea_Blog_1024x1024.jpg
image size	1024 × 1024
source site	heavenlytealeaves.com

The alt text usually does not describe every object in the image.

OBELICS: processing real web pages



Original webpage, current rendering

Record

0 text: “The hike is a 5-day hike through wild and pure ocean fronts and tropical forests.”



1 image
alt: G-_Photoshop_IMG_2615



2 image
alt: IMG_2599

3 text: next paragraph about trail difficulty

MINT-1T: a real PDF record

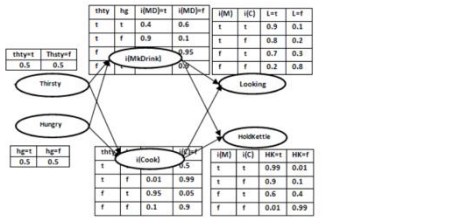


Figure 1: Elders Collective Intentions CBN. There are two top nodes, $Thirsty$ and $Hungry$, belonging to the pre-intentional level describing the causes that might give rise to the intentions nodes, $i(MkDrink)$ and $i(Cook)$. Two action nodes, $Looking$ and $HoldKettle$, are observable. They are children of the intention nodes.

Let $s_i, 1 \leq i \leq n$, be the first action of agent a_i in P. Determine indices $d_i, 1 \leq i \leq n$, such that $p_{d_i} = s_i$. Group the agents having the same first action, i.e. with the same index d_i. They are doing the same task or at least some part together. Intuitively, these agents have a joint parallel subplan. Suppose we obtain m groups $g_1, ..., g_m$, and g_t is responsible for the subplan $[p_{j_t+1}, ..., p_{j_{t+1}}]$ where $1 \leq t \leq m$ and $0 = j_1 < ... < j_{m+1} = k$. It is easily seen that the grouping is unique for a given set of agents and a given plan.

In order to check if there are expectation actions that reflect agents' mutual expectations, we consider two cases: mutual expectations between agents in a group and between agents in consecutive or subsequent subplans groups [2]. This way, the number of interactions amongst the agents that need to be observed (e.g. by an activity recognition system) is considerably reduced. Furthermore, the number of possible expectation actions between two particular agents is reduced and being more specified.

Expectations inside Group When some agents are assigned or intended to do something together, they expect from each other to do the task. Thus, the expectation actions can be a "complain" action when a group member "deviates" from the task without an "inform" action.

An example where two people have an intention to walk together (Tomasello 2008), if one of them changes his/her direction without informing the other, he/she obviously will be complained on by the other. This is to differentiate from the situation where those two people walk in step by chance.

In short, we say a group of agents having the same subplan (or part of it) actually work cooperatively if we observe expectation action "complain" if there is any agent deviating from the assigned task.

When agents cooperatively work together, if some agent cannot manage his task, we usually observe some kind of "help" action. However, we believe that kind of action is mostly preceded by a "complain" action. Inability to do the assigned task is also one kind of deviation.

The term "deviation" is used here as a general term, and should be specified for concrete application domains. For example, in spatial domain (Sukthankar 2007; Devaney and Ram 1998) an agent is said to deviate if he is not in his assigned position or does not move in the right direction or does so with wrong velocity.

Expectations between Consecutive Groups When agents are in consecutive groups, says g_t and g_{t+1}, responsible for the subplans $[p_{j_t+1}, ..., p_{j_{t+1}}]$ and $[p_{j_{t+1}+1}, ..., p_{j_{t+2}}]$, respectively, agents from the first would expect some result from the second, who in turn expect the agents from the first to use their result. If the agents from one of the groups "deviated" from their task or did not finish it as assigned, the agents from the other group would "complain". For example, two people want to make coffee, one is assigned to boil water and other is assigned to look for the coffee. Having boiled the water, the first may have the "expect result" action "ask for the coffee", or he may have "complain" action if the second person could not find the coffee after a while. Some "help" action may follow, but we could conclude the collective intentionality solely with the "complain" action. However, if the second found the coffee earlier, he has an "expect use" action, e.g. "this is the coffee, use it!".

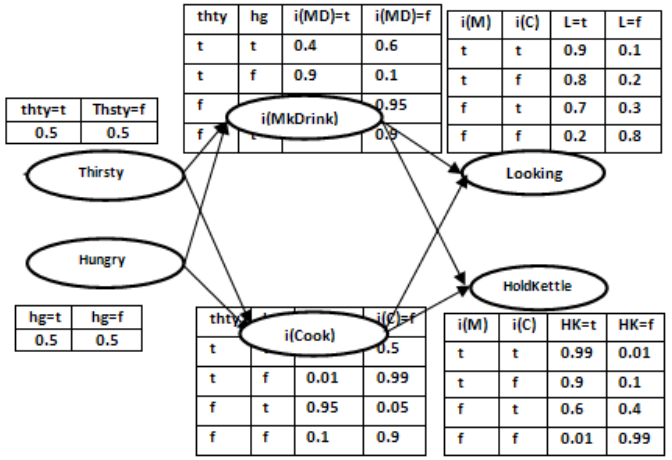
Let $result_t$ be the assigned result of group g_t. Usually, this result comes from the last action, $p_{j_{t+1}}$, of the group subplan. For IR, we assume that the set of possible actions reflecting the expectation of $result_t$ of the action $p_{j_{t+1}}$, denoted by $expect_result_t$, and the set of possible actions reflecting the expectation of using $result_t$, denoted by $expect_use_t$, are given.

Then, we say these two groups of agents are working towards achieving a collective intention if either agents in g_{t+1} have some action belonging to $expect_result_t$, or agents in g_t have some action belonging to $expect_use_t$, or there are "complain" actions from one of the groups. As long as the

[2] It might be the case that one group interacts (produces or consumes results) with more than one other group, but that is not considered in this paper.

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PDF page 4



Released sequence

...text before figure
image: page_3_image_33
... Figure 1: ...

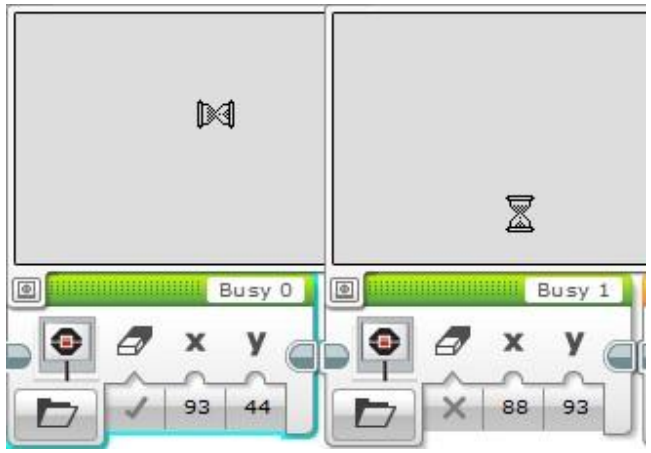
Actual stored fields

Field	Value
page	3
xref	33
pixels	434 × 302

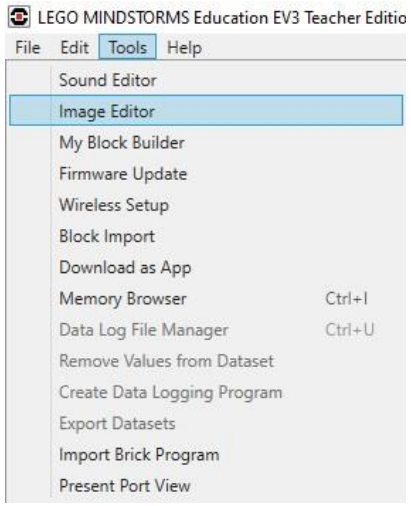
xref = PDF object ID

A real record can lose the main text

Released OBELICS sample: an EV3 programming page



Original alt: “code”



“howtoaccess”

Retained text (excerpt)

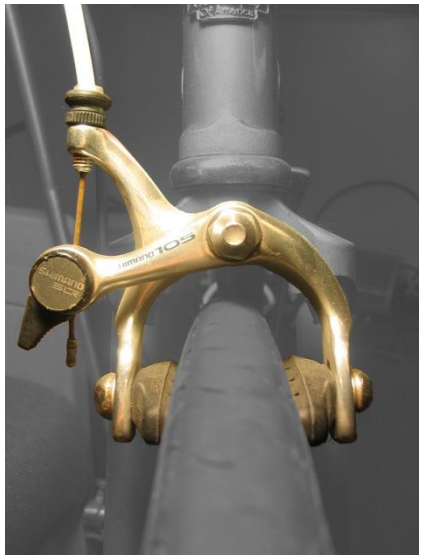
“EV3 Brick Error Symbols?”
“Why does display output affects motors?”

Released field	Contents
images	6 images
texts	Only related-question links

The images survived. The main explanation did not. Inspect records after extraction.

Page structure is also important

Two brake designs on Wikipedia's "Bicycle brake" page



A: "Single pivot side-pull caliper brake"



B: "Low profile 'traditional' cantilever brake"

Fields in the page's HTML

Field	A	B
alt description	-	-
figure caption	Caption A	Caption B
linked image	image1	image2

Alt text misses both captions
We need to rely on the structure of the page

2 Alignment and filtering



	Methods in this stage
Is the image usable?	Size and logo rules
Which text belongs to it?	CLIP scores and MMC4 global matching
Which pairs should remain?	Score thresholds and Data Filtering Networks

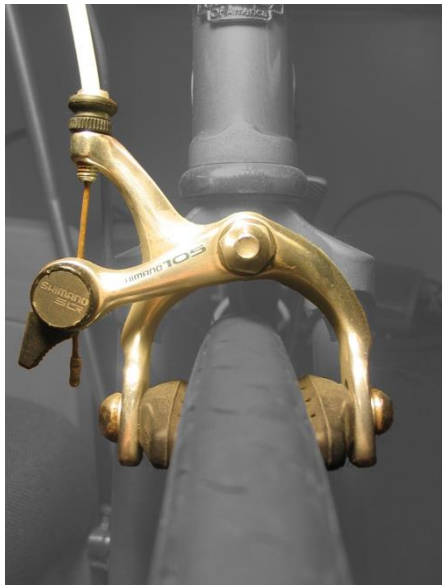
Filtering removes small images and logos

Example rule: keep article images whose shorter side has at least 200 pixels (px)

Image	Image from the page	Width × height	Decision
	A: caliper original	1536 × 2048	Keep
	A': an added copy of A	1536 × 2048	Keep for now
	A-small: a small version for the page	250 × 333	Keep for now
	B: cantilever original	972 × 1069	Keep
	C: small spoon-brake image	120 × 90	Drop: too small
	Site logo	40 × 54	Drop: a logo, not an article image

Wikipedia contributors, Bicycle brake Images: Aezram / Keithonearth and AndrewDressel Credits and links in notes

Four candidate texts for the same two images



A: caliper photo



B: cantilever photo

T1 Single pivot side-pull caliper brake.

T2 Low profile ‘traditional’ cantilever brake.

T3 A bicycle brake reduces the speed of a bicycle

T4 Tools

CLIP scores every image–text pair

$$s(I,T) = \frac{f(I)^{\top} g(T)}{\| f(I) \| \| g(T) \|}$$

f(I) and g(T): CLIP image and text embeddings. Scores are cosine similarity

Image	T1: caliper	T2: cantilever	T3: opening text	T4: Tools
A	0.2784	**0.2889**	0.2715	0.1735
B	**0.2960**	0.2692	0.2779	0.1778

OpenAI CLIP ViT-L/14, 224-pixel center crop, CPU float32

A cutoff of 0.30 rejects both original image–caption pairs

MMC4 filters images, then matches them to text

1 Keep each image whose highest text score is at least 0.15

A: 0.2889 B: 0.2960 Both pass

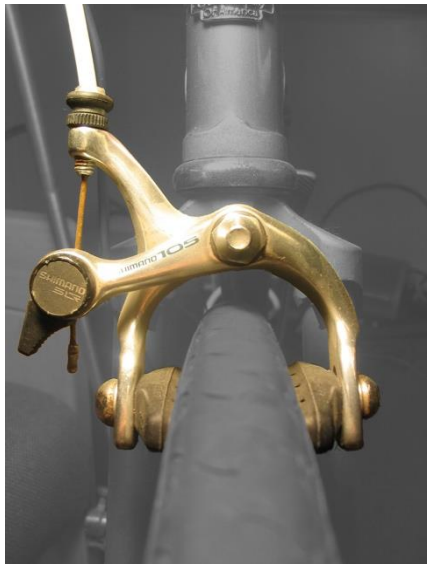
$$\max_{x} \sum_{i,j} s_{ij}\, x_{ij} \quad \text{s.t.} \quad \sum_{j} x_{ij} = 1, \quad \sum_{i} x_{ij} \leq 1, \quad x_{ij} \in \{0,1\}$$

2 Maximize total score. Use each image once and each text at most once

Choice of pairs	Total cosine score
A–T1 and B–T2	0.5476
A–T2 and B–T1	0.5848

s_{ij} = pair score. x_{ij} = 1 selects image i with text j

The highest-scoring pairs may have the wrong captions



A → T2
Score 0.2889

Assigned text describes a cantilever brake



B → T1
Score 0.2960

Assigned text describes a side-pull caliper brake

Both images prefer the same caption

To isolate the conflict, keep only T1 and T3 from the measured scores



A: caliper



B: cantilever

Image	T1: caliper	T3: opening text
A	**0.2784**	0.2715
B	**0.2960**	0.2779

T1 Single pivot side-pull caliper brake.

T3 A bicycle brake reduces the speed of a bicycle

Per-row argmax assigns both images to T1

Both images pass the 0.15 filter. A one-to-one match must give T1 to only one image

Global matching chooses the best total score

Assignment rule	A receives	B receives	Total score
Best score in each row	T1	T1	0.5744*
Greedy: choose for A first	T1	T3	0.5563
Global: choose both together	T3	T1	0.5675

* This total uses T1 twice, which the one-to-one rule forbids

A loses 0.0069 if it gets T3

$$0.2784 - 0.2715 = 0.0069$$

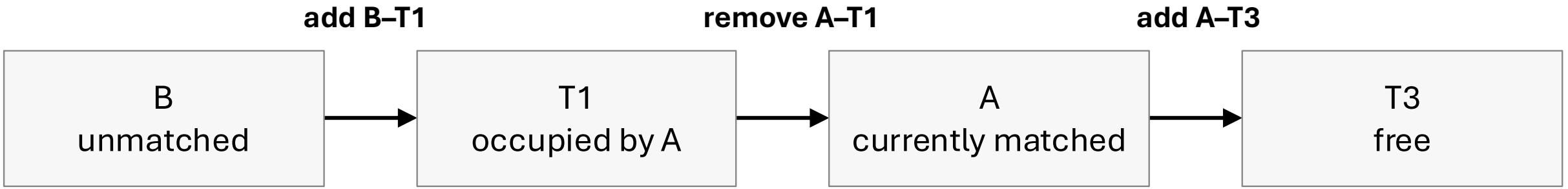
B loses 0.0181 if it gets T3

$$0.2960 - 0.2779 = 0.0181$$

Give T1 to B and T3 to A. The valid total is 0.0112 higher than choosing for A first

An augmenting path resolves the caption conflict

An augmenting path adds and removes pairs to match one more image



Start with A–T1. Compare two ways to give B a caption

Direct: add B–T3

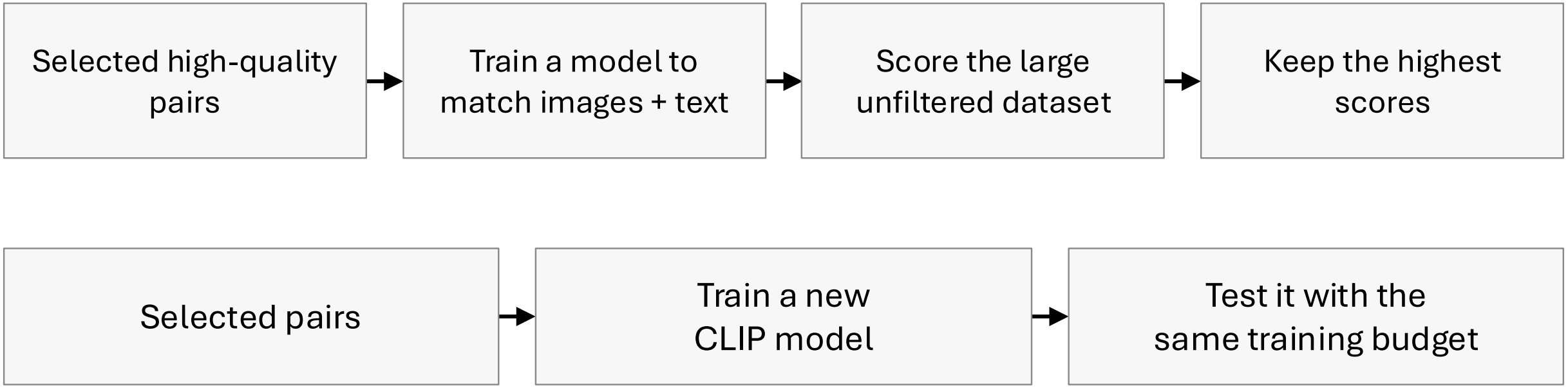
$$+0.2779$$

Reassign: add B–T1, remove A–T1, add A–T3

$$0.2960 - 0.2784 + 0.2715 = 0.2891$$

Choose the larger increase: move A to T3 and give T1 to B

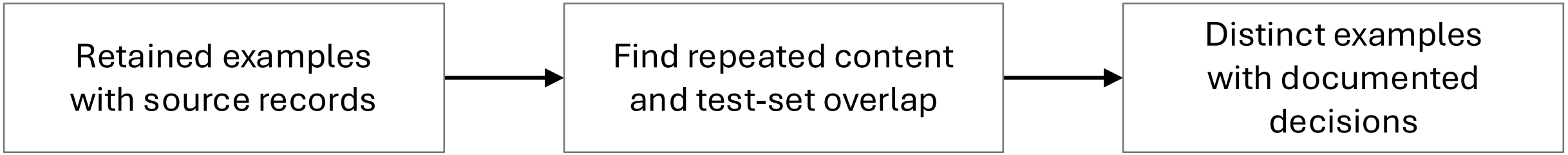
Data Filtering Networks learn which pairs to keep



The filter learns from a smaller set of selected pairs

Evaluate the new CLIP model. Check whether the filter drops useful unfamiliar examples

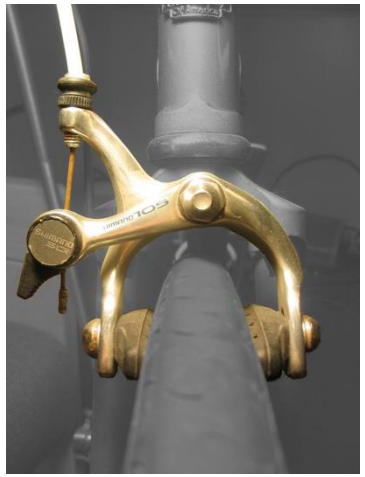
3 Duplicate removal



Comparison	**Methods in this stage**
Files / image copies	File hashes and perceptual image hashes
Overlapping text	Bloom filters, MinHash and LSH
Similar content	Feature-based deduplication and source audits

Awadalla et al. MINT-1T Abbas et al. SemDeDup Gadre et al. DataComp

Deduplication removes repeated images



A: original
File hash: f09b5a7d…



A': repeated download
File hash: f09b5a7d…



A-small: thumbnail
File hash: e552a017…



B: different brake
File hash: fd927af2…

SHA-256 finds byte-identical copies: remove A'

dHash compares visual patterns: A vs A-small differs by 2 bits, A vs B by 35 bits

At distance ≤ 4, keep the largest copy. Output: (A, caption A) and (B, caption B)

Wikipedia contributors, Bicycle brake Images: Aezram / Keithonearth and AndrewDressel Credits and links in notes

Bloom filters test whether text was seen before

Hash each word sequence to bit positions, then set those bits to 1

Position	0	1	2	3	4	5	6	7
Bit	0	1	1	0	1	0	1	0

Insert A: set positions {1, 4} to 1
Insert B: set positions {2, 6} to 1

Query C: {1, 6} are 1, so it may be present
Query D: position 3 is 0, so it is absent

Text deduplication

A shingle is a consecutive word sequence. Here we use 3 words

ID	3-word shingle	ID	3-word shingle	ID	3-word shingle
1	a bicycle brake	9	bicycle or prevents	16	bicycle brake needs
2	bicycle brake reduces	10	or prevents the	17	brake needs regular
3	brake reduces the	11	prevents the wheels	18	needs regular inspection
4	reduces the speed	12	the wheels from	19	regular inspection before
5	the speed of	13	wheels from moving	20	inspection before every
6	speed of a	14	from moving when	21	before every ride
7	of a bicycle	15	moving when parked		
8	a bicycle or				

A contains IDs 1–13 B contains IDs 1–15

$$J(A,B) = \frac{|A \cap B|}{|A \cup B|} = \frac{13}{15} \approx 0.867$$

Jaccard = shared / total unique
C contains ID 1 and IDs 16–21

Broder, On the Resemblance and Containment of Documents Leskovec et al., Mining of Massive Datasets, Ch. 3

MinHash samples shared shingles

Use the same random order for every document

Compute the first shingle ID found in each order

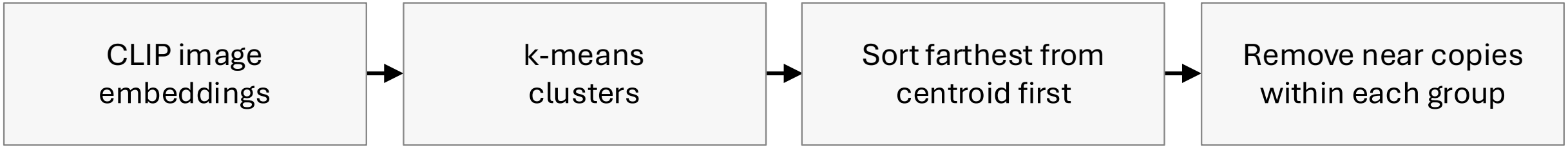
Order	Beginning of random shingle order	A	B	C
1	15, 8, 5, 13, 19, …	8	15	19
2	10, 19, 4, …	10	10	19
3	1, 16, 15, …	1	1	1
4	1, 16, 6, …	1	1	1
5	19, 2, 7, …	2	2	19
6	7, 1, 16, …	7	7	1

A and B match in 5 of 6 positions: estimated similarity = 0.833

$$P[\text{same winning shingle}] = J(A, B)$$

Exact value: 0.867
More orders reduce noise

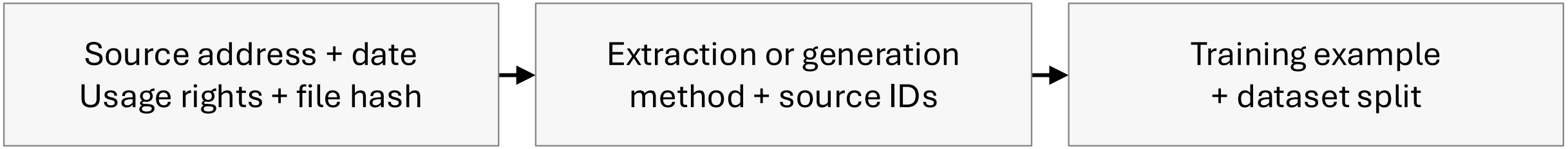
SemDeDup removes images with similar features



Processing order (toy scores)	**Highest cosine similarity to earlier images**	**Keep if score < 0.95?**
A: brake photograph	No earlier point	Yes
A-small: resized version of A	0.99	No
B: different brake mechanism	0.84	Yes

Image-only deduplication can remove useful differences in captions

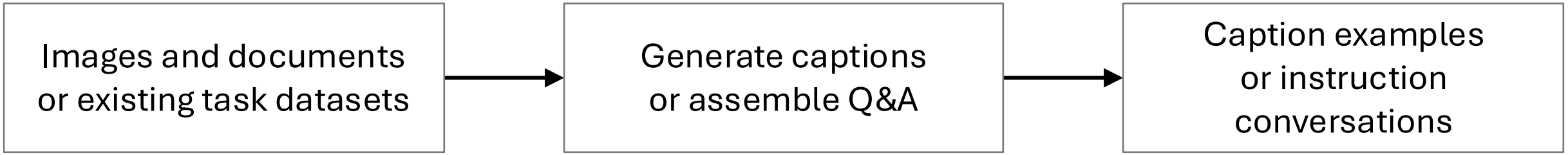
Tracking sources, privacy, and test-data leakage



Operation	**Concrete check**
Detect sensitive content	Scan personal data + inspect images
Group duplicates	Compare images, text, and sources
Track generated examples	Store model and source IDs
Check test-data leakage	Match test images and edited copies

Schuhmann et al. LAION-5B: An Open Large-Scale Dataset for Training Next Generation Image-Text Models | Gadre et al. DataComp: In Search of the Next Generation of Multimodal Datasets | Awadalla et al. MINT-1T: Scaling Open-Source Multimodal Data by 10x: A Multimodal Dataset with One Trillion Tokens

4 Additional training supervision



Need	**Example in this stage**
More descriptive text	BLIP replaces weak web captions
Instruction data	LLaVA generates questions and answers
Visual task coverage	FineVision combines instructions in existing datasets

New or merged examples return to quality, duplicate and overlap checks

Li et al. BLIP Liu et al. Visual Instruction Tuning Wiedmann et al. FineVision

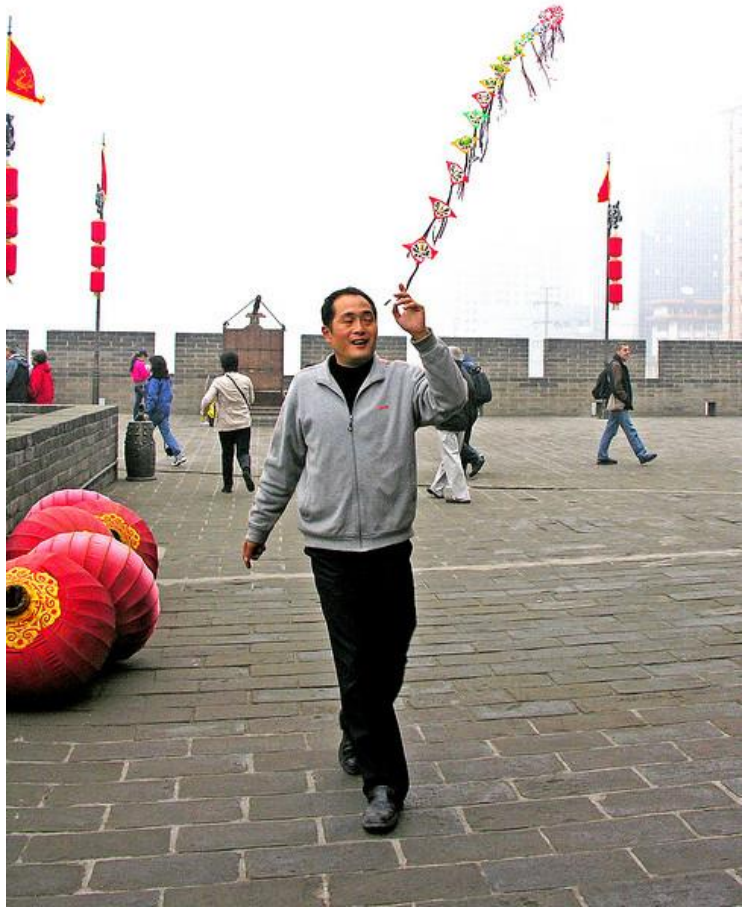
BLIP: a real caption replacement



Source	Actual text	Filter decision
Web text	living in my little white house	Reject
BLIP caption	a tiny white flower with a bee in it	Keep

Captioner proposes a new description.
Then use a image–text matching filter to check.

LLaVA: a real instruction record



User turn

“What do you see happening in this image?”

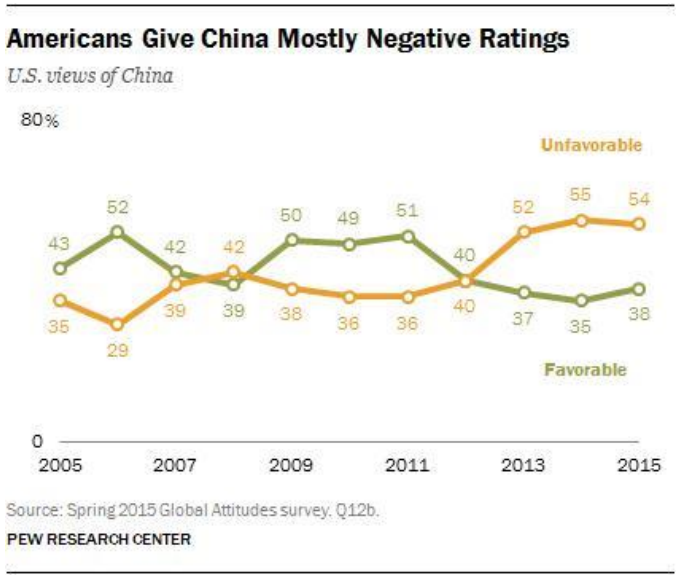
Assistant turn (excerpt)

“The kite has multiple sections attached to it”

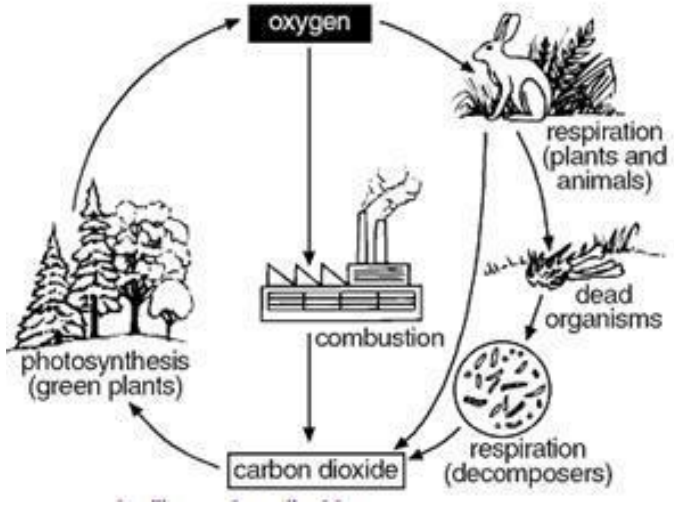
Original LLaVA generation: GPT-4 reads COCO captions and boxes, then writes instruction turns.

FineVision collects many kinds of visual tasks

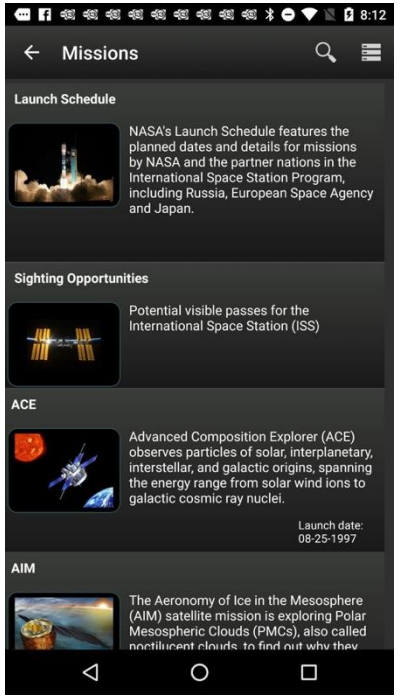
200+ sources, 185 subsets. Converted into instruction conversations.



ChartQA: read a chart



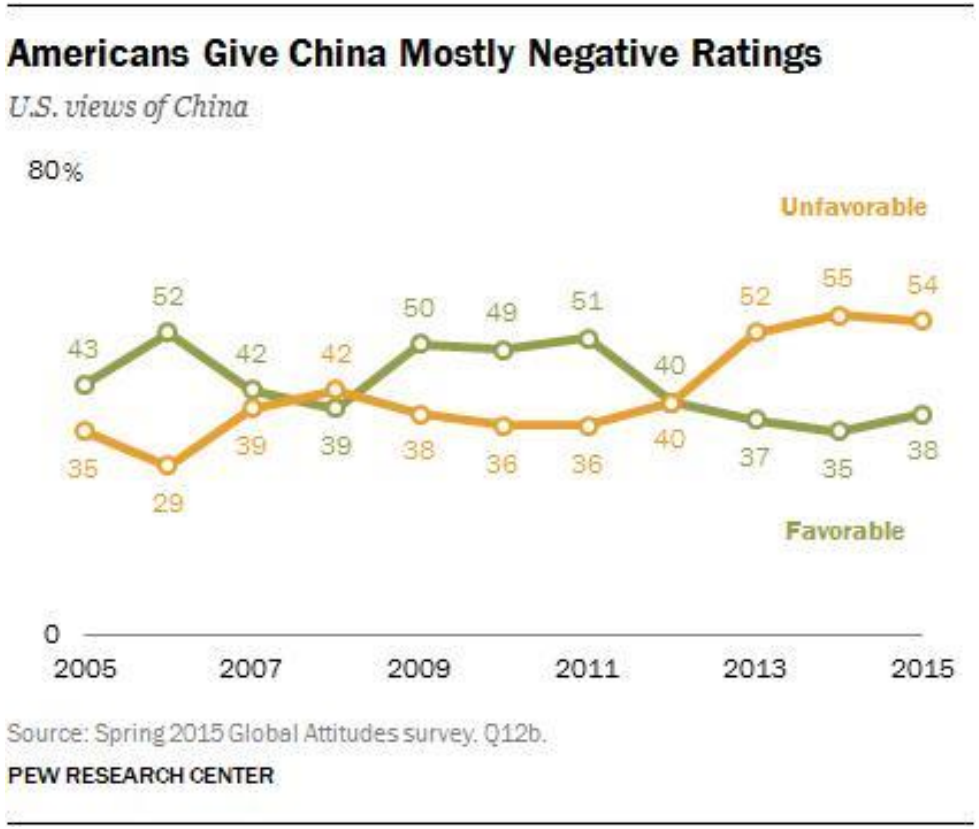
AI2D: interpret a diagram



ScreenQA: read a screen

17.3M images 24.3M samples 88.9M turns 9.5B answer tokens

FineVision: Merge multiple queries into a single document



Turn	Question and answer
1	Is the value of Favorable 38 in 2015? Yes.
2	How many values are below 40 in Unfavorable graph? 6.

Each turn reuses the same image.

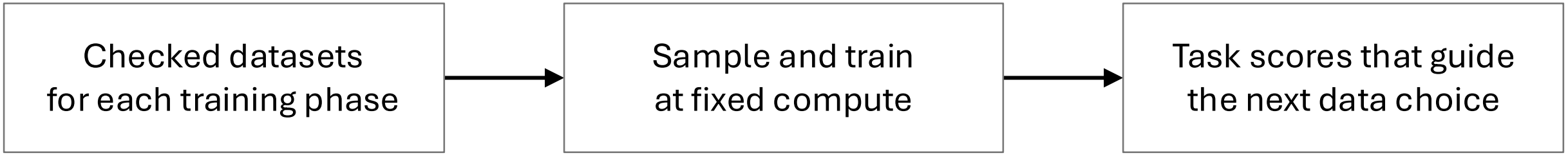
FineVision: duplicate detection

SSCD is an image encoder trained to detect copies.
Flag a pair when cosine similarity ≥ 0.95.

Where to compare	**What FineVision does**
Within training sources	Inspect near copies; merge overlapping sources. Three AI2D variants become ai2d_merged.
Across different benchmarks	Flag likely image overlap.

Choose whether to merge questions, remove a sample, or keep distinct supervision.

5 Mixtures, training and evaluation



Decision	**Evidence in this stage**
Sampling shares	Mixture shares and repeated passes through data
Batch selection	JEST jointly selects examples
Other comparison	DataComp: training budget; MM1: model and data choices

Evaluate the repair assistant, then revise sources, filters or sampling shares

Evans et al. JEST Gadre et al. DataComp McKinzie et al. MM1

Sampling

Source (toy)	Unique tokens	Sampling fraction	Average passes
Image captions	20B	40%	2×
Interleaved documents	30B	40%	1.33×
Text-only	100B	20%	0.2×

E_i = average passes, p_i = source i's share, D = total tokens, U_i = distinct source tokens
Example: D = 100B. A 40% share of a 20B source gives 40B / 20B = 2 passes

Next: JEST chooses examples within a batch

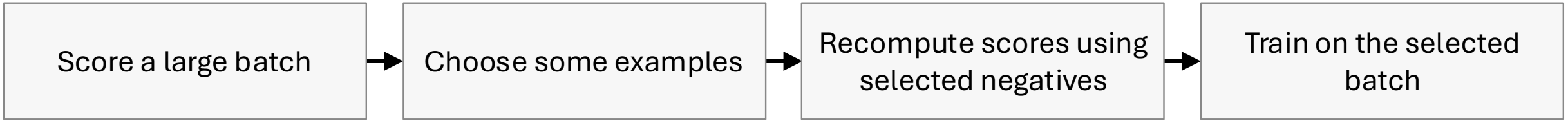
JEST: Use a pretrained model on human-filtered data

B: candidate batch

$$S(B) = \mathcal{L}_{\text{learner}}(B) - \mathcal{L}_{\text{reference}}(B)$$

Learner: model being trained. Reference: pretrained on curated data

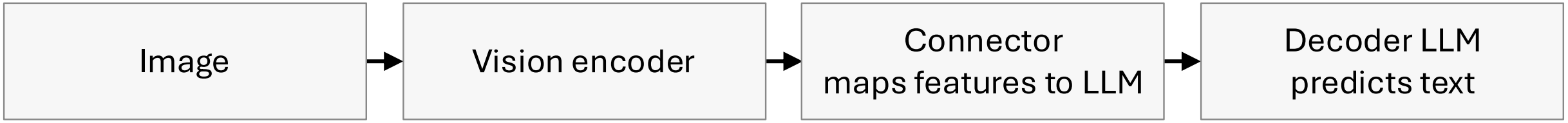
Example (toy losses)	Learner loss	Reference loss	Difference
A: reference knows it, learner does not	2.0	0.2	1.8
B: both models find it hard	2.0	1.9	0.1
C: both models find it easy	0.3	0.1	0.2



We can achieve faster training for models trained on bigger datasets

Evans et al. Data Curation via Joint Example Selection Further Accelerates Multimodal Learning

MM1: encoder size and image resolution

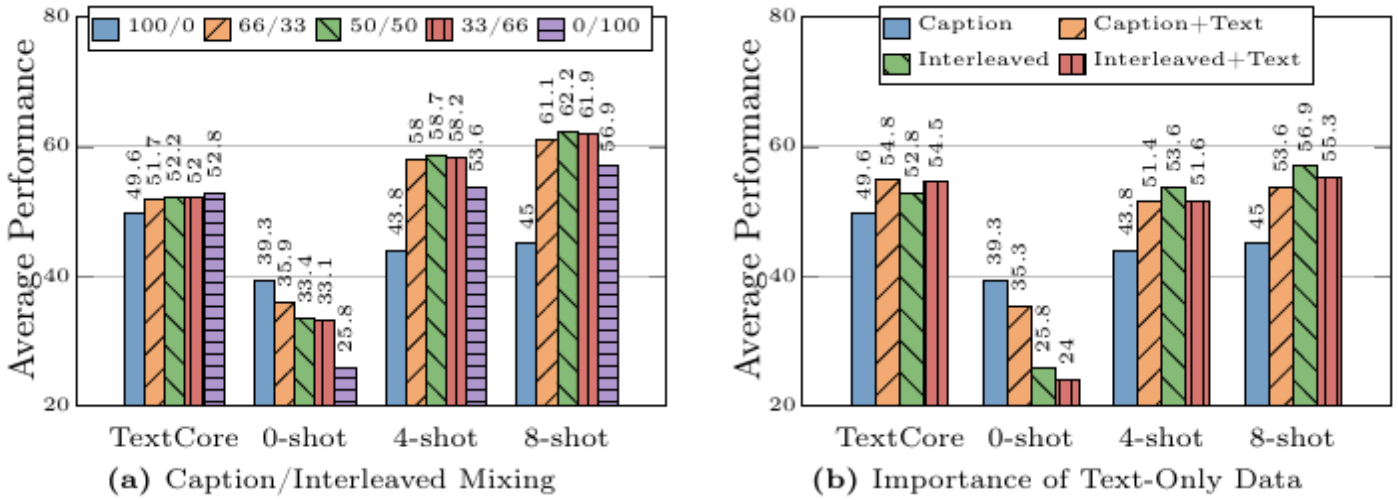


Mean score on 8 captioning and visual question answering (VQA) tasks

Image encoder / image size	**0-shot**	**4-shot**	**8-shot**
CLIP ViT-L, 224 px	36.9	58.7	62.2
CLIP ViT-L, 336 px	39.9	62.4	66.0
CLIP ViT-H, 336 px	40.5	62.6	66.3

MM1 Table 1: 2.9B LLM. Encoder data: DFN-5B + VeCap synthetic captions
Increasing resolution helps more than switching from ViT-L to ViT-H in this comparison

MM1 data types help different tasks



Caption data improves zero-shot performance

Interleaved documents improve few-shot performance

Text-only data helps preserve language performance

MM1's chosen mixture
45% caption
45% interleaved
10% text-only

Original Fig. 5a–b. TextCore summarizes performance on language tasks

The data pipeline determines what the model learns

Stage
1 Collect and extract
2 Align and filter
3 Deduplicate and audit
4 Add supervision
5 Mix, train and evaluate

2 Scaling laws

Chinchilla: trade off model size and training tokens

Mixed-modal models: measure whether sharing helps or hurts

Chinchilla: choosing model size and token count

For the same training compute, is a larger model always better?

Design choice	Experimental setting
Model type	Dense autoregressive language models
Training data	MassiveText: English web pages, books, news, code, and reference text
Training runs	400+ runs with about 70M–16B+ parameters and 5B–500B tokens
Measured outcome	Prediction loss after training. Lower is better

Compute-optimal means the lowest loss for a fixed training-compute budget

Scaling-law quantities and loss

Quantity	Meaning	What must stay comparable
N	Model parameters	Same model family and counting rule
D	Training tokens processed	Same tokenizer and data sources
C	Training FLOPs	Same forward/backward cost convention
L	Held-out cross-entropy	Same evaluation set and normalization

Keep the data, tokenizer, and model family comparable when fitting the law

Chinchilla fits loss from model size and data size

$$L(N, D) = E + \frac{A}{N^{\alpha}} + \frac{B}{D^{\beta}}, \qquad C \approx 6ND$$

E: loss floor. A and B: error scales. α and β: rates of improvement

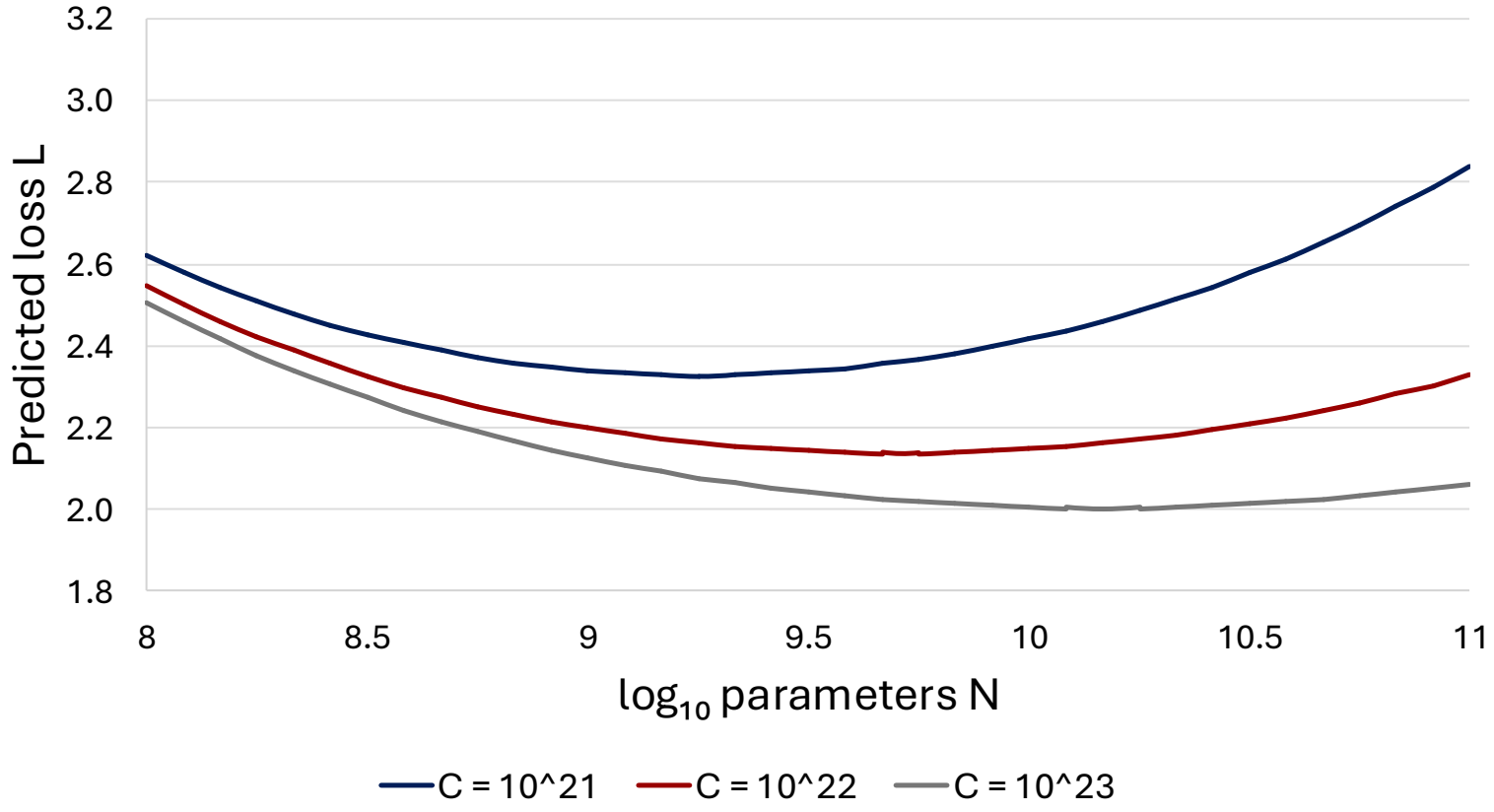
$$\alpha \frac{A}{N^{\alpha}} = \beta \frac{B}{D^{\beta}}$$

$$N_* \propto C^{\frac{\beta}{\alpha+\beta}}, \qquad D_* \propto C^{\frac{\alpha}{\alpha+\beta}}$$

N* and D* minimize loss at compute C. Similar α and β imply similar growth in model size and tokens

Hoffmann et al. Training Compute-Optimal Large Language Models

IsoFLOP curves compare models at equal compute



Small model:
too few parameters

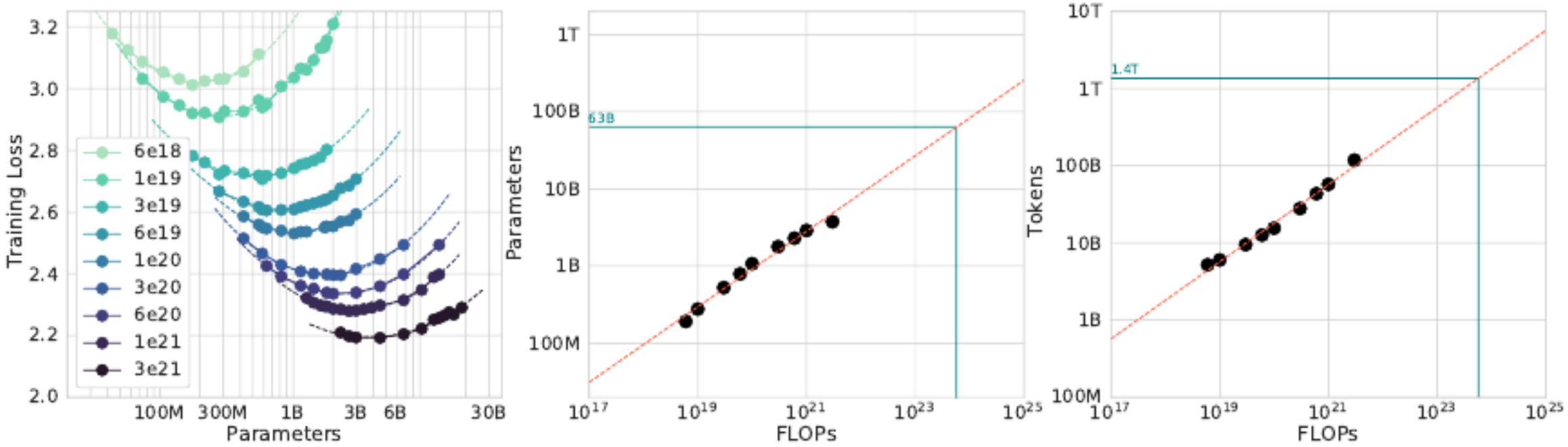
Large model:
too few training tokens

Chinchilla: 70B, 1.4T tokens
Gopher: 280B, 300B tokens

Each curve fixes compute and varies N and D. Curves are calculated from the fitted equation

Hoffmann et al. Training Compute-Optimal Large Language Models

Chinchilla Fig. 3: finding the lowest-loss model



Left: each curve has one compute budget	**Middle and right: plot each curve's minimum**
x: parameters y: final loss	x: FLOPs y: best model size or token count
Moving right trades tokens for parameters	Fit power laws to these optimal allocations
The minimum gives the best N at this budget	Extend the fitted trend to Gopher's budget

Chinchilla tests the prediction with a larger run

Model	Parameters	Training tokens	MMLU 5-shot
Gopher	280B	300B	60.0%
Chinchilla	70B	1.4T	67.5%

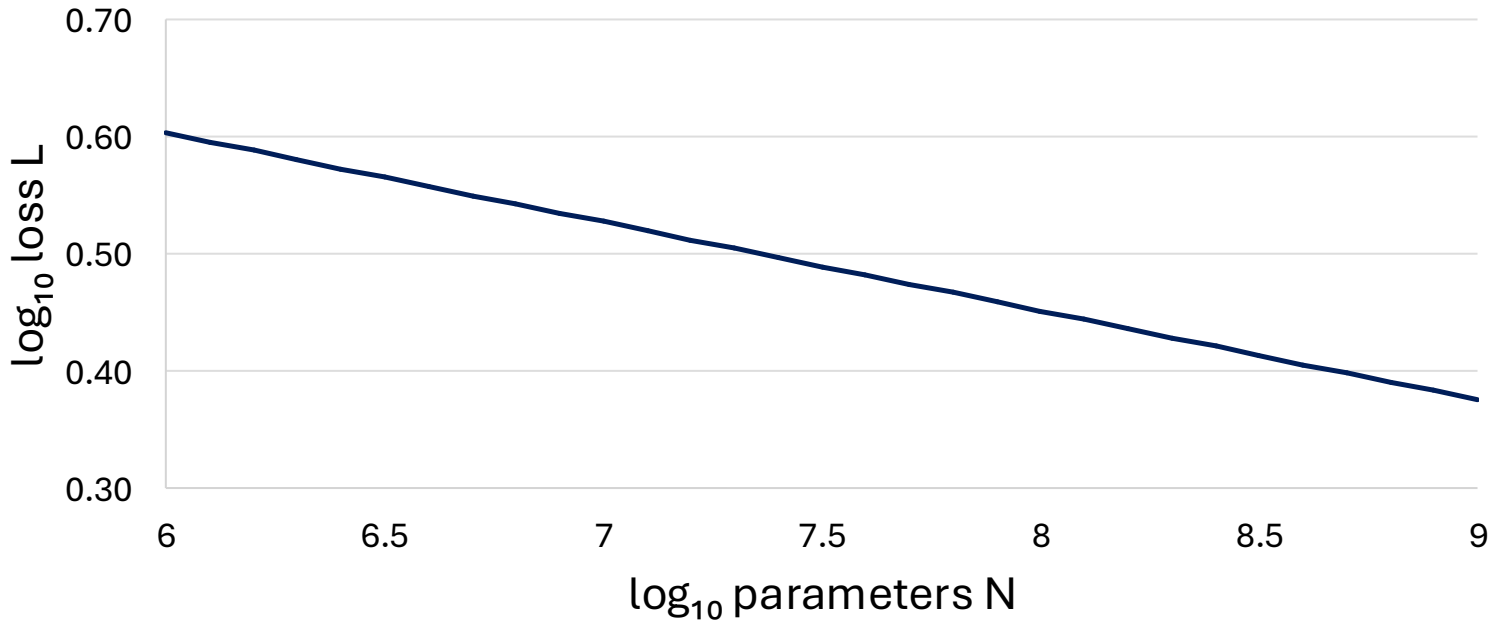
Chinchilla uses about the same training compute as Gopher, with more tokens and fewer parameters

The larger run supports the prediction fitted from smaller runs

The optimizer, tokenizer, and MassiveText sampling also changed

Kaplan: loss follows a power law

$$L(N) = \left(\frac{N_c}{N}\right)^{0.076}$$



N_c is a fitted scale constant. L is loss

On log–log axes, slope = −0.076

10× more parameters gives about 0.84× loss

With data as the limit: exponent ≈ 0.095

Published model-size fit: enough data and training so model size is the main limit

Kaplan et al. Scaling Laws for Neural Language Models

Multimodal training adds a new data-mixture decision

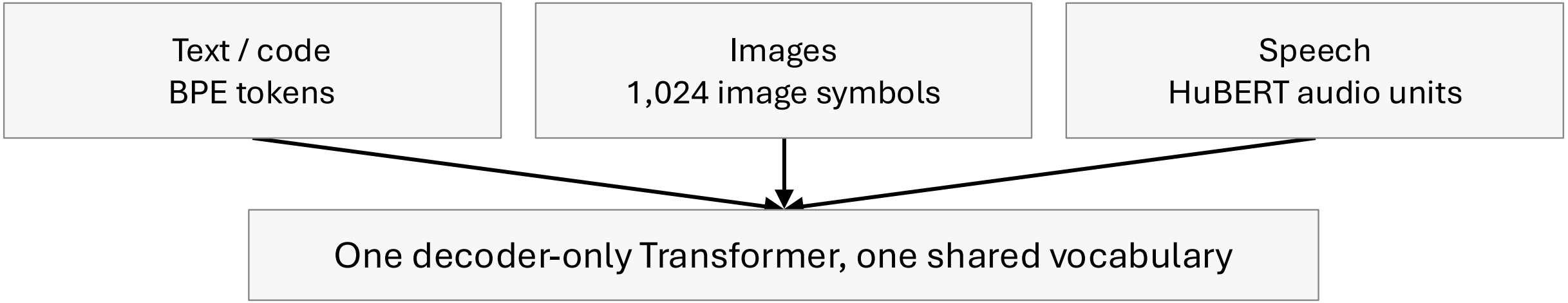
Chinchilla keeps the data setup fixed	**A mixed-modal model also needs a mixture**
Choose parameter count N	Choose which modalities share one model
Choose total training tokens D	Choose each dataset’s share of D
Fit loss as a function of N and D	Measure whether sharing helps or hurts
Test the predicted allocation at larger scale	Test whether the shared model beats separate models

Aghajanyan et al. test pairs of datasets. Each dataset supplies 50% of the training tokens

Hoffmann et al. Training Compute-Optimal Large Language Models Aghajanyan et al. Scaling Laws for Generative Mixed-Modal Language Models

Aghajanyan et al.: when does sharing help?

Can one model predict tokens better than two separate models?



250+ runs, 7 dataset types, 8M–6.7B parameters for fitting

The paper tests its prediction at 30B parameters.

The paper’s seven training datasets

Dataset label	Data source	What a training example contains
Text	OPT language-model dataset, 180B tokens	Natural-language text
Image	600M images from LAION and a separate web crawl	Image symbols with captions removed
Image-Text	The same image pool with available captions	Image tokens plus paired caption text
Speech	Web podcasts/news + public speech corpora	Audio symbols without transcripts
Speech-Text	MLS and VoxPopuli speech/transcript datasets	Aligned speech and transcript
Code	InCoder data	Source-code text
Molecules	ZINC chemical database, written as SMILES strings	SMILES: text symbols for atoms and bonds

How images, speech, and text become tokens

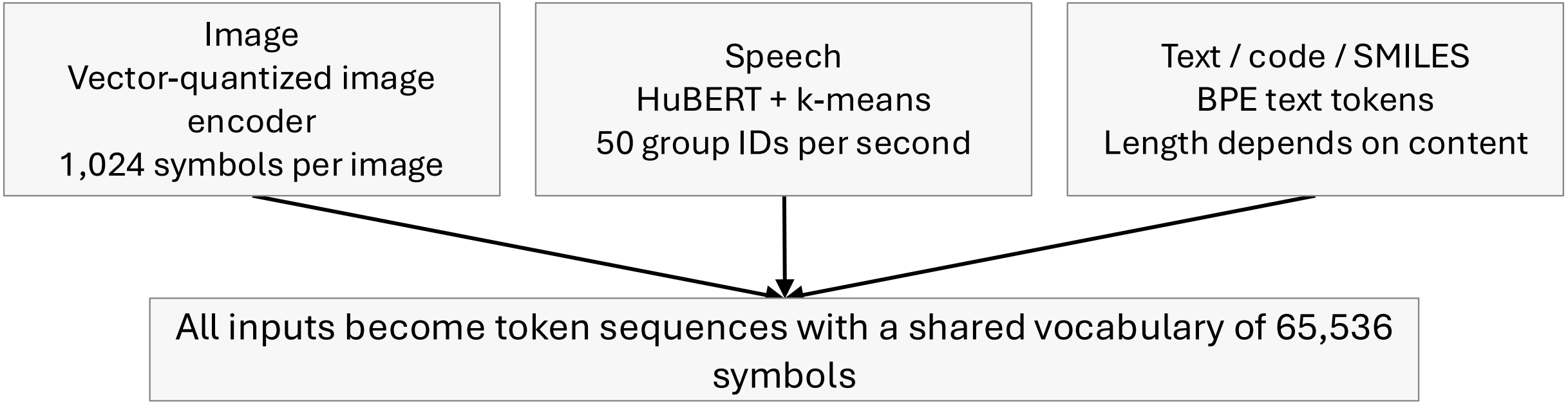


Image codebook: 8,192 entries. Speech codebook: 2,000 entries

Training runs used to fit and test the law

Variable	Values or control
Model sizes used for fitting	8M, 125M, 350M, 760M, 1.3B, 2.7B, 6.7B
Training-token budgets	5B, 10B, 100B
Datasets	7 individual datasets and 7 selected dataset pairs
Mixture for each pair	50% of tokens from each component dataset
Prediction task	Decoder-only Transformer with causal masking/infilling
Data repetition	Single epoch: each training document appears once
Test beyond the fitted sizes	30B Speech/Text model, trained on 50B tokens

Fit models up to 6.7B parameters. Test the prediction with a 30B model

Fitting the law separately for each dataset

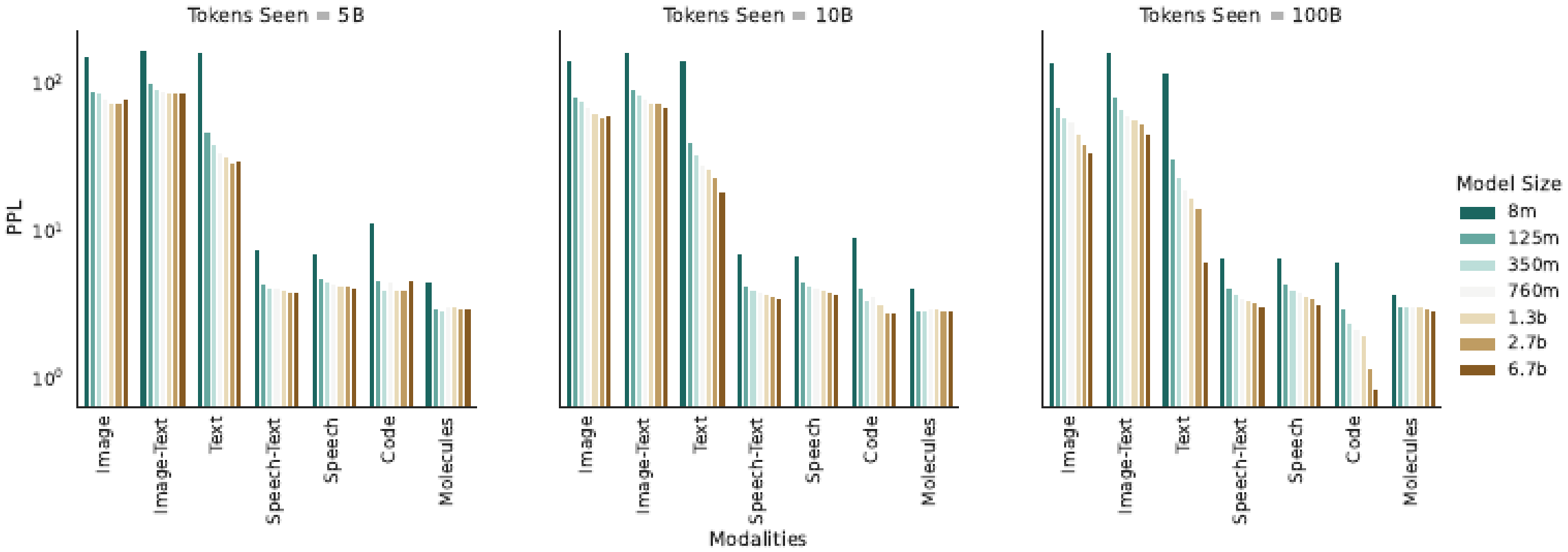
$$L_j(N, D_j) = E_j + \frac{A_j}{N^{\alpha_j}} + \frac{B_j}{D_j^{\beta_j}}$$

j labels a dataset. D_j counts its tokens. E_j is its loss floor. A_j and B_j set error size

Dataset	α: model-size exponent	β: token-count exponent
Text	0.18	0.22
Image	0.13	0.13
Speech	0.31	0.24
Code	0.37	0.32

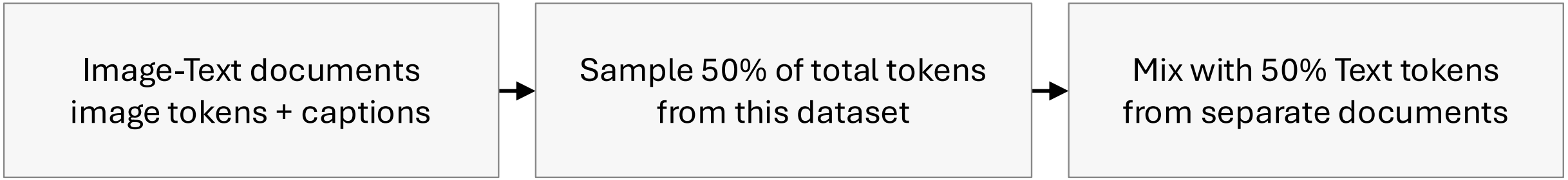
Fit five coefficients using Huber loss and multiple L-BFGS initializations

Fig. 2: larger models usually have lower loss



Panels: token budgets. Colors: model sizes. y: the paper's PPL on a log scale

Counting image tokens in a 50:50 dataset mixture



Comparing a shared model with separate models

Model trained only on dataset i N parameters, D_i tokens	Model trained only on dataset j N parameters, D_j tokens	One model trained on N parameters, D_i + D_j tokens

$$L_{\text{base}} = \frac{L_i(N, D_i) + L_j\left(N, D_j\right)}{2}, \qquad D_i = D_j$$

$$\Delta_{ij} = L_{\text{mixed}} - L_{\text{base}}$$

Δ > 0: competition (higher loss). Δ < 0: synergy (lower loss)

The mixed-modal law predicts sharing costs and gains

$$L_{ij} = \frac{L_i(N, D_i) + L_j\left(N, D_j\right)}{2} + \Delta_{ij}$$

$$\Delta_{ij} = -C_{ij} + \frac{A_{ij}}{N^{\alpha_{ij}}} + \frac{B_{ij}}{\left(D_i + D_j\right)^{\beta_{ij}}}$$

Term in Δ	**Meaning**
1 Sharing benefit	Benefit predicted at very large N and D
2 Model-size penalty	Extra error from limited model size
3 Data-size penalty	Extra error from limited training data

i and j identify the dataset pair. C_{ij} is the sharing benefit. Plain C earlier counted training compute

How much data/compute do we need to see the synergy

$$C_{ij} = \frac{A_{ij}}{N^{\alpha_{ij}}} + \frac{B_{ij}}{D^{\beta_{ij}}}, \qquad D = D_i + D_j$$

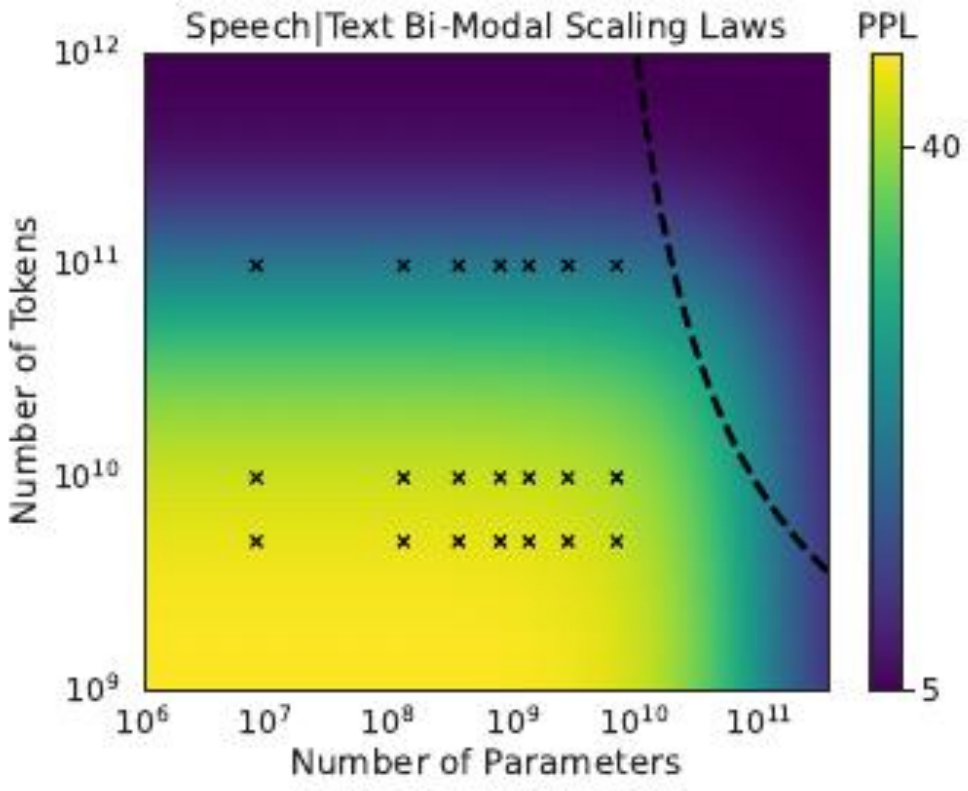
At the barrier, Δ = 0: the shared and separate models have equal loss

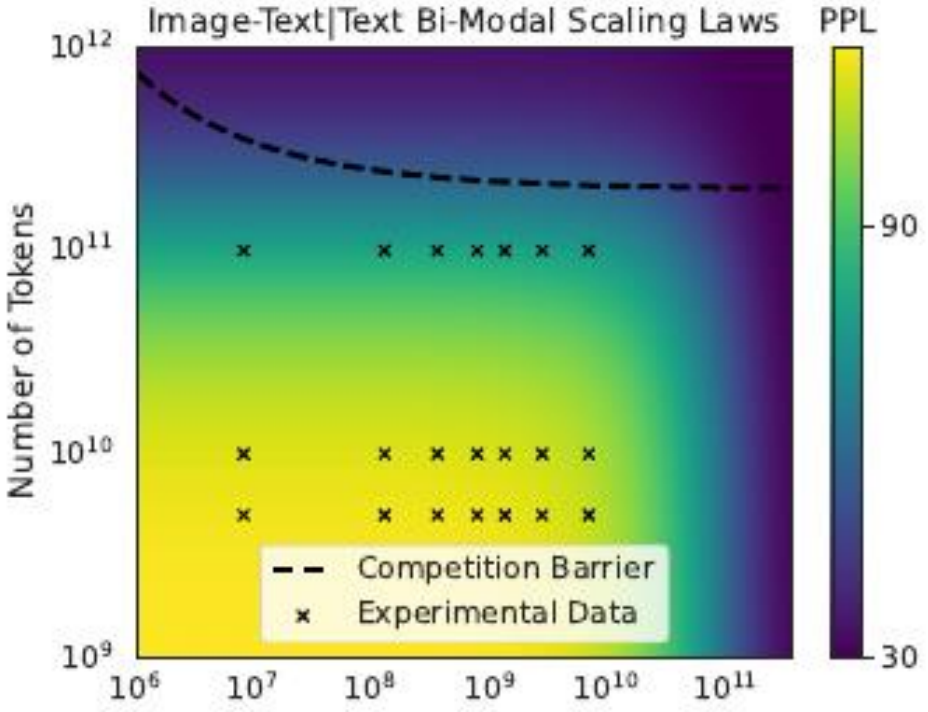
$$D_{\text{barrier}}(N) = \left(\frac{B_{ij}}{C_{ij} - A_{ij}/N^{\alpha_{ij}}} \right)^{1/\beta_{ij}}$$

Along this boundary, choose N and D with the lowest compute 6ND

Speech/Text prediction: 28.35B parameters and 45.12B tokens

Fig. 4: different pairs have different boundaries





x: parameters N. y: total tokens D. Both use log axes

Color: predicted PPL. Dashed line: $\Delta = 0$. Crosses: measured model/data sizes

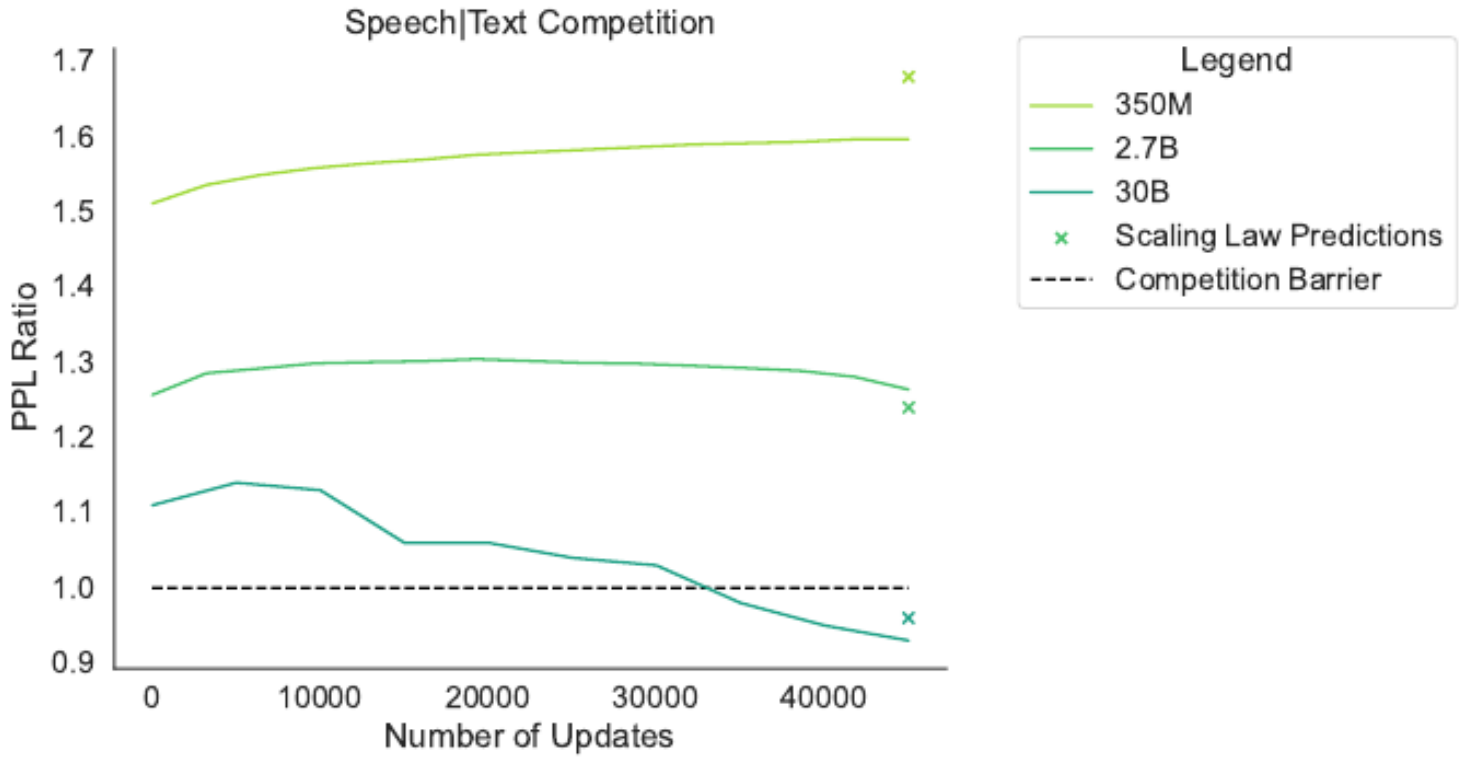
Above and to the right of the line, the fit predicts lower loss from sharing

Testing the Speech/Text prediction at 30B parameters

Step	What the paper does
Fit	Fit the sharing boundary using smaller Speech/Text models
Choose a target	Lowest predicted compute on the boundary: 28.35B parameters, 45.12B tokens
Run the test	Train a nearby 30B model on 50B tokens
Compare	Train separate and shared models at 350M, 2.7B, and 30B parameters
Read the result	Compare shared and separate losses during training

The test measures token prediction. It does not test how well the model follows instructions

Fig. 5: the 30B model crosses the predicted boundary



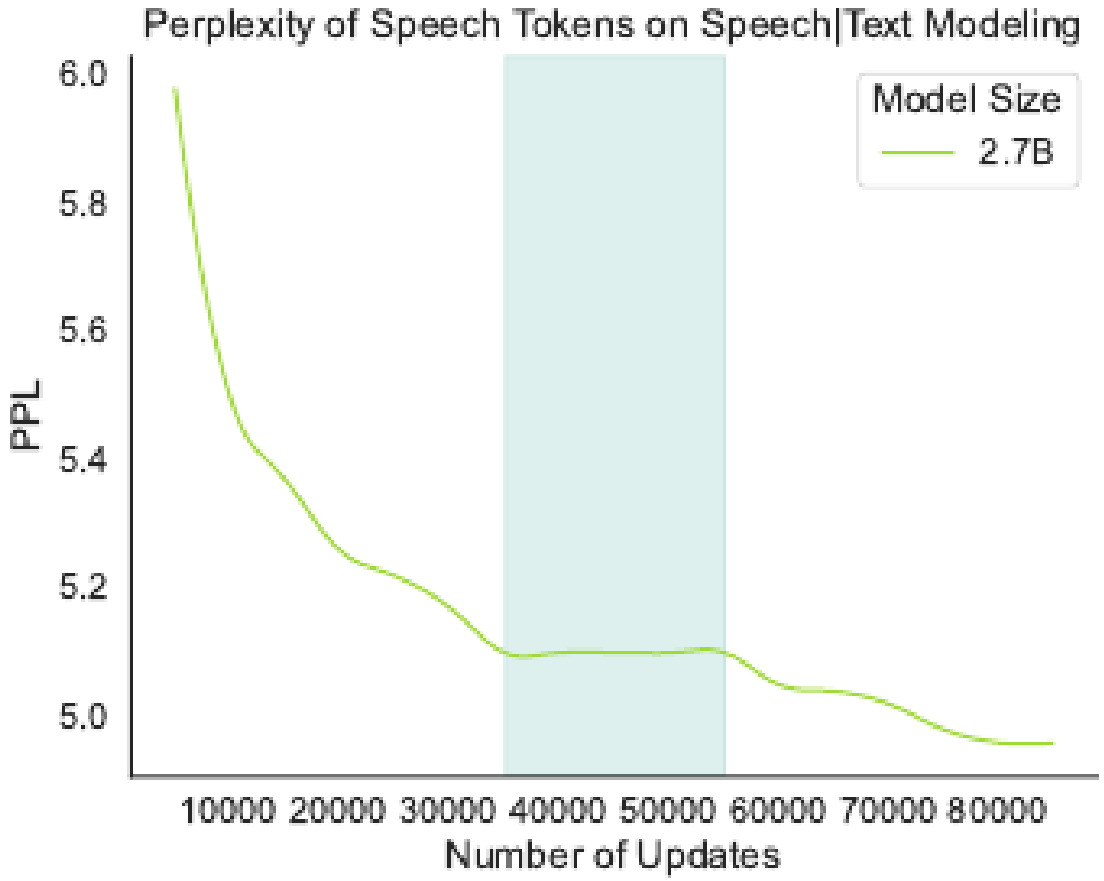
x: training updates
y: ratio of perplexity scores
Colors: 350M, 2.7B, 30B

Dashed line at 1
Line at 1 means equal scores
Crosses show the predictions

The 30B curve drops below 1
The smaller-model curves stay above 1

Source inconsistency: the printed ratio and the caption disagree on which model is in the numerator

Fig. 6: speech learning can pause during joint training



Model: 2.7B Speech/Text
Score: perplexity on speech tokens

x: training updates
y: speech-token perplexity

Speech perplexity barely changes for about 15,000 updates

Limitation: Cross-modality, Mixture study

Questions for Discussion

- If training data is filtered by previous-generation models, can new models learn things those models do not understand?
- In particular, what evidence would convince you that multimodal training leads to new capabilities?
- More questions you might have!